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Tracking Surface Changes Caused by Volcanic Activity

Chapter G of
Recommended Capabilities and Instrumentation for Volcano Monitoring in the United States

Scientific Investigations Report 2024–5062

U.S. Department of the Interior
U.S. Geological Survey

Cover. Photograph from the northwest rim of Kīlauea, Hawaiʻi, of the resumption of lava lake activity within Halemaʻumaʻu following a new eruption on September 29, 2021. The photograph was taken shortly after the eruption began and shows two web cameras and volcanic fissures producing lava that is accumulating within the crater. This was the second summit eruption after the devastating 2018 caldera collapse and lasted over a year, until December 9, 2022. Photograph by L. Gallant, U.S. Geological Survey, September 30, 2021. Background image shows a typical view of the 2008–2018 lava lake in the Overlook crater within Halemaʻumaʻu, Kīlauea. Photograph taken from a helicopter by Tim Orr, U.S. Geological Survey, on August 16, 2013.

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Conversion Factors

International System of Units to U.S. customary units

Multiply	By	To obtain
	Length	
meter (m)	3.281	foot (ft)
meter (m)	1.094	yard (yd)
kilometer (km)	0.6214	mile (mi)

Abbreviations

ASTER	Advanced Spaceborne Thermal Emission and Reflection Radiometer
AVHRR	Advanced Very High Resolution Radiometer
COSMO-SkyMEd	Constellation of Small Satellites for Mediterranean basin Observation
DEM	digital elevation model
FLIR	forward-looking infrared
GLISTIN-A	Airborne Glacier and Land Ice Surface Topography Interferometer
GOES	Geostationary Operational Environmental Satellite
InSAR	interferometric synthetic aperture radar
lidar	light detection and ranging
MODIS	Moderate Resolution Imaging Spectroradiometer
NASA	National Aeronautics and Space Administration
SAR	synthetic aperture radar
SfM	structure from motion
SWIR	short-wave infrared imagery
UAS	unoccupied aircraft system
USGS	U.S. Geological Survey
VIIRS	Visible Infrared Imaging Radiometer Suite

Chapter G

Tracking Surface Changes Caused by Volcanic Activity

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Introduction

Dynamic volcanic landscapes produce various changes at the surface of volcanic edifices. For example, rising magma can induce thermal emissions, formation of ground cracks, and variations in glacier and edifice morphology; volcanic deposits from eruptions can transform the land surface with tephra fall, pyroclastic flows, lava flows and domes, and lahars; and geomorphic changes from landslides and lahars can occur in the absence of unrest or eruption.

The best way to detect these changes is with imagery obtained via satellite, aircraft (including unoccupied aircraft systems, or UAS), and ground-based imaging. Rapid advances in imaging technologies have been leveraged by the U.S. Geological Survey (USGS) Volcano Hazards Program to improve the ability to monitor volcanoes. To this end, the guidance outlined here provides a framework for tracking volcanic unrest and the emplacement and evolution of volcanic deposits, further elucidating the processes associated with volcanic eruptions. The techniques currently used include (1) various telemetered and non-telemetered cameras, (2) high-resolution ground-based optical (visible to short-wave infrared wavelengths) and thermal infrared photography, (3) satellite and airborne thermal, optical, and synthetic aperture radar (SAR) imagery, and (4) light detection and ranging (lidar) surveys from airborne and ground-based platforms. Given that similar or overlapping techniques are applied to meet the capabilities listed in this chapter, we first provide an overview of remote sensing techniques. The use of UAS in monitoring surface change is briefly mentioned in this chapter and described in more detail in the dedicated UAS chapter (chapter L, this volume; Diefenbach, 2024).

Instrumentation

Space-Based Sensors

Satellite remote sensing provides synoptic views of a volcanic edifice and the surface changes and emplacement of deposits through time before, during, and after an eruption. High-resolution satellite imaging with visible, infrared, and SAR sensors is critical for mapping and characterization of volcanic vents and deposits over large areas and inaccessible or dangerous terrain (Patrick and others, 2016; Waythomas and others, 2020). Satellite data sources vary widely with respect to spectral coverage (wavelengths of 300 nanometers to 300 millimeters; that is, ultraviolet, visible, middle infrared, thermal infrared, and microwave), spatial resolution (submeter to tens of kilometers), and temporal resolution (as often as 1 minute to several weeks between repeat acquisitions). In contrast to airborne and ground-based remote sensing, space-based remote sensing imagery can capture local (tens of kilometers) to regional (thousands of kilometers) scales, typically with global coverage. New satellite sensors and constellations of satellites with multiple platforms operating the same sensor, like the Constellation of Small Satellites for Mediterranean basin Observation (COSMO-SkyMed) and Sentinel-2 A and B, offer greater spatial, spectral (additional channels and increased sensitivity), and temporal resolution. This allows for observations of steam, ash, and high-temperature thermal emissions, as well as surficial change from deformation and eruption deposits. Advances in lower spatial resolution meteorological satellites, including the launch of Visible Infrared Imaging Radiometer Suite (VIIRS) instruments on a constellation of satellites starting in 2011 and new Geostationary Operational Environmental Satellite (GOES) satellites, are providing visible and infrared views at higher temporal (as short as 1-minute for GOES mesoscale) and spatial resolution (as high as 375 meters [m] VIIRS imagery bands) than ever before to capture large-scale volcanic activity in near-real time. Other developments include faster transfer of large datasets, cloud-based data storage and processing, and automated processing of near-real-time data that can be programmed to provide alerts when noteworthy changes are detected.

Collaboration among the USGS, the National Aeronautics and Space Administration (NASA), and other agencies and research groups worldwide have helped bring research and experimental operational products into use by the USGS for volcano monitoring. For example, with the launch of Landsat 8 in 2013 and the launch of Landsat 9 in 2021, the USGS Landsat program provides routine observations of the Earth's surface. The Landsat 8 and 9 sensors provide better spatial and spectral resolution than their predecessors, with bands covering 0.43–12.51 micrometer, spatial resolution as high as 15 m, and a repeat time of 16 days. Complementary sensors aboard the European Space Agency Sentinel-2 satellite constellation are also operational (repeat interval of 5 days), and together these instruments provide regular multispectral imaging of volcanoes, eruption deposits, high-temperature thermal features, vegetation changes, and other phenomena.

Although traditionally used in volcanology for deformation monitoring (and highlighted in chapter D, this volume, on ground deformation and gravity change; Montgomery-Brown and others, 2024), SAR data provide the unique capability to view the surfaces of volcanoes through clouds, steam, and ash emissions, revealing new eruption deposits (for example, lava flows, dome growth, ash fallout, and lahars) and other surficial changes (such as ground cracking), and are also used to generate digital elevation models (DEMs). The USGS has used various SAR data from cooperating agencies such as the European Space Agency (Sentinel), Canadian Space Agency (Radarsat), German Aerospace Center (TerraSAR-X and TanDEM-X), Japan Aerospace Exploration Agency (Advanced Land Observing Satellite), and Italian Space Agency (COSMO-SkyMed). These datasets offer resolutions less than 1 m and repeat cycles of days to weeks. The joint NASA-Indian Space Research Organization Synthetic Aperture Radar Mission, with a planned launch date in 2024, will mark the first satellite-based SAR mission by the United States and will provide routine data for volcano monitoring.

Airborne Sensors

Technological developments in camera systems, UAS platforms, and image-processing software have revolutionized the resolution and quality of airborne remote sensing data products. The data generated may include visible images, thermal infrared images acquired with forward-looking infrared (FLIR) cameras, topographic data produced by lidar and structure from motion (SfM) methods, and airborne SAR. On-demand overflights with crewed and unoccupied aircraft can provide regular observations of activity to track eruption evolution and detect changes, thereby improving eruption forecasts (Poland and others, 2016; Neal and others, 2019; Dietterich and others, 2021). Optical and thermal imagery and video are used for observations and characterization of eruptive activity, such as eruptive style, deposit extents, eruption rates, and lava temperature. Data acquired from airborne surveys can be processed with SfM photogrammetric methods to construct DEMs at centimeter resolution (for example, Dietterich and others, 2021) and high-resolution orthoimagery, including visible wavelengths for mapping and thermal infrared wavelengths for tracking surface temperatures (for example, Patrick and others, 2017).

Airborne SAR has been widely used to acquire high-resolution DEMs at U.S. volcanoes (for example, the USGS Alaska Mapping Initiative) and has been deployed in eruption response to document topographic change during eruptions. NASA's Airborne Glacier and Land Ice Surface Topography Interferometer (GLISTIN-A) SAR instrument flies on a crewed aircraft and utilizes single-pass interferometric synthetic aperture radar (InSAR) to construct DEMs on demand (for example, Lundgren and others, 2019). Similarly, airborne lidar is deployed to collect high-resolution, high-precision terrain data at volcanoes before, during, and after eruptions. Lidar data, particularly acquired from low-altitude aircraft, offer centimeter-scale resolution and the ability to filter results to obtain bare-earth topographic data in vegetated regions. Unlike space-based remote sensing, acquisition of airborne remote sensing data is, in most cases, spatially limited to individual volcanoes.

Ground-Based Sensors

Ground-based optical and thermal cameras, including both telemetered and non-telemetered systems, are a proven means of detecting eruption onset, tracking and documenting eruptive activity, and monitoring morphological and temperature changes at volcanoes (for example, Major and others, 2008). Their use depends on adequate visibility and lighting (for optical systems), both of which can be limited by weather conditions, viewing geometry, and darkness, such as during winter eruptions at high-latitude volcanoes in Alaska. Low-light (near-infrared) and thermal infrared cameras can readily capture warm features and work overnight and through minor steam and gas emissions. Current web cameras (fig. G1) can capture images with as high as 4,000-pixel resolution with high dynamic range, representing a vast improvement in imaging quality compared to web cameras from the early 2000s. Some camera systems offer remote control of pan, tilt, and zoom functions to change their field of view, streaming or time-lapse video acquisition, and (or) capture and storage of video on-board for later retrieval. Many readily available modern camera systems even use cellular telemetry and thus permit rapid deployment outside of an existing telemetry network if there is cell reception.

Individual cameras, including high-speed cameras, have provided quantitative information on lava fountain heights (Wolfe and others, 1988; Orr and others, 2015), the elevations of lava lakes (Orr and Rea, 2012; Patrick and others, 2014),

Figure G1. Photograph showing the Mount Cleveland web camera at Concord Point, Chuginadak Island, Islands of Four Mountains, Aleutian Islands, Alaska. The station also hosts the CLCO seismic station and an array of infrasound sensors. The south flank of little-studied Tanax̂ Angunax̂ (Tana) volcano rises in the background. Photograph by M. Kaufman, University of Alaska Fairbanks, July 19, 2018; used with permission.

lava lake surface motion (Patrick and Orr, 2018), spattering dynamics (Mintz and others, 2021), and explosion particle tracking (Taddeucci and others, 2012; Vanderkluysen and others, 2012). Ground-based cameras deployed in combination can be used to quantify topographic changes (for example, Major and others, 2009). Other types of ground-based remote sensing instrumentation, such as SAR and automated laser rangefinders, have also been used to characterize surface change (for example, Di Traglia and others, 2014; Patrick and others, 2019), but require favorable viewing conditions for deployment.

Recommended Capabilities

Collect Baseline Imagery and Topography

Volcano observatories assess volcanic hazards to support land-use planning, emergency-response, and public awareness. This requires studies to determine the style and frequency of past eruptions and the potential effects of future activity. Fundamental to such studies are aerial and satellite images to identify past eruptive products and accurately map their distribution. In addition, DEMs are required for hazard assessment and modeling because many volcanic hazards are controlled by topography. For example, assessing the regions that are most at risk from inundation by lahars, pyroclastic density currents, and lava flows requires accurate and up-to-date DEMs. Similarly, terrain is a fundamental control on edifice stability, and DEMs are therefore needed for slope-stability assessment. Finally, preeruptive imagery and DEMs are key baseline data for comparing with syneruptive or posteruptive data to detect and characterize eruptive activity and volcanic unrest.

Instrumentation

The most used image sources for geologic and hazard mapping include high-resolution aerial and satellite images. High-spatial-resolution (submeter), cloud-free visible and near-infrared imagery of quiescent volcanoes is widely available. These data are supplemented by lower resolution (tens of meters) Landsat or Sentinel data for additional spectral information, such as for identifying the extent of hydrothermal alteration. Airborne SAR instruments, like NASA's GLISTIN-A, and airborne lidar have been deployed to collect high-resolution and high-precision terrain data at volcanoes before, during, and after eruptions (Lundgren and others, 2019; Mosbrucker and others, 2020).

Recommendations

We suggest acquisition of moderate-resolution baseline imagery (for example, Landsat 8 or 9 or Sentinel-2) and 10-meter minimum resolution DEMs at all levels 1 and 2 volcanoes. For level 3 volcanoes, we suggest acquisition of a 5-meter minimum resolution DEM and high-spatial-resolution satellite images (WorldView or equivalent; 1 meter or better resolution) or stereo aerial imagery to allow for detailed geologic and hazard mapping. The same procedures would ideally be followed for level 4 volcanoes except that the DEMs should have a 1-meter minimum resolution to improve hazards modeling of lahars, pyroclastic density currents, lava flows, and other phenomena. A high-spatial-resolution orthoimage would ideally be re-collected for levels 3 and 4 volcanoes at least every year, and DEMs would be re-collected at least every 10 years. High-resolution imagery and DEMs would ideally be collected regularly during and following eruptions for volcanoes of all levels. The frequency of acquisitions are typically tailored to the hazards, eruptive style, timescales, weather, and remoteness of eruptions. All DEMs should be bare earth, representing the topography with vegetation and structures removed, and all imagery should be georeferenced and orthorectified.

Track Unrest and Eruptive Activity

Restless and higher threat volcanoes require routine observational monitoring to maintain awareness. When these volcanoes enter a state of unrest or erupt, there is then a constant need to observe and document activity, map deposits, and measure eruption rates. Various techniques have been developed and utilized that are cost effective and provide increased viewing potential, as well as increased safety for scientific field crews. These techniques include (1) telemetered and non-telemetered ground-based cameras (including time-lapse, low-light, and infrared), (2) aerial observation and imaging using crewed aircraft and UAS, (3) high-resolution satellite optical and thermal images, (4) airborne and satellite SAR, and (5) ground and airborne lidar. Owing to the many variables regarding access, logistics, eruption style, and climate, the use of these techniques needs to be tailored to the individual volcano. For example, radar (and, in the case of hot targets, infrared) techniques are particularly useful where visual observations are limited by steam, clouds, or fog, which can complicate interpretations of other monitoring data.

Instrumentation

Imagery from all sensors at all spatial and temporal scales is critical to detecting changes associated with volcanic unrest and the deposits and dynamics of eruptions. These images can be analyzed qualitatively to detect and describe changes in surface features (for example, appearance or disappearance of a lake, a new lahar, or an ash deposit). High-resolution imagery for ground, airborne, and satellite platforms can also be orthorectified and analyzed for mapping and measuring these features (for example, mapping the extent of a lava flow through time). Analysis uses spectral characteristics, feature tracking, and change detection to identify, trace, and quantify surface deposits and changes at volcanoes. During the 2019–20 Shishaldin Volcano eruption, for example, thermal, optical, and SAR satellite imagery were jointly analyzed to document the changes in the summit crater, lava flows, and plumes associated with eruption to inform assessment of current activity and potential hazards (fig. G2).

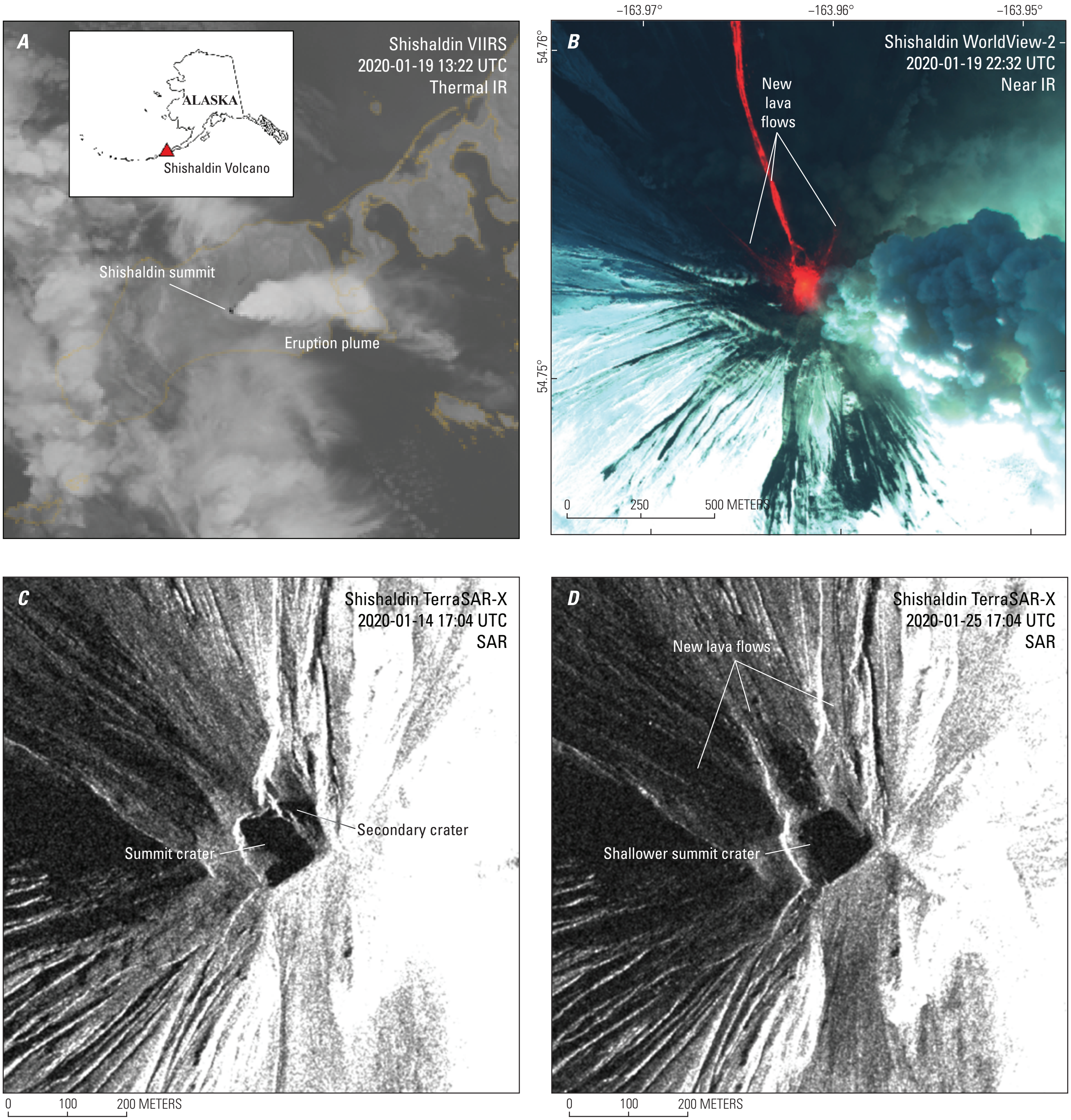
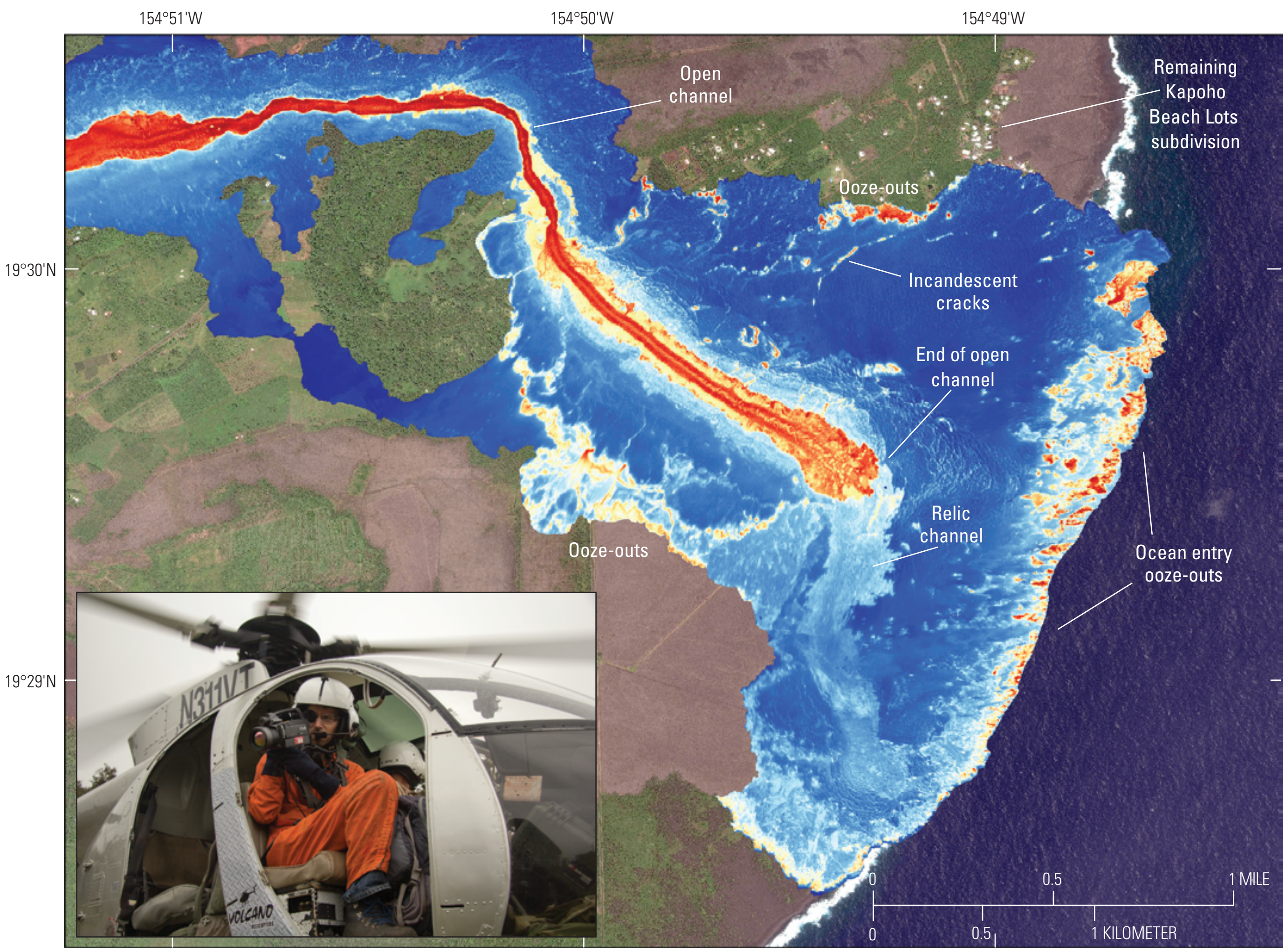


Figure G2. Satellite imagery of the January 19–20, 2020, explosive event of Shishaldin Volcano, Alaska. *A*, Thermal infrared (IR) image (Visible Infrared Imaging Radiometer Suite [VIIRS] on January 19 at 13:22 coordinated universal time [UTC]) of the diffuse ash and gas plume produced by lava fountaining. *B*, Off-nadir near-infrared false color image (WorldView-2 on January 19 at 22:33 UTC) of lava flows to the north, northwest, and northeast being fed from a central scoria cone that is actively fountaining and producing a vertical column of steam and minor ash. *C*, Preeruptive (January 14) high-resolution TerraSAR-X synthetic aperture radar (SAR) backscatter image showing the summit region of Shishaldin Volcano (illumination from the southeast), including a deep summit crater with a secondary crater to the northeast. *D*, Posteruptive (January 25) high-resolution TerraSAR-X backscatter image showing the summit region of Shishaldin Volcano (illumination from the southeast), including new lava flows visible in part *B* and a single, shallower summit crater. WorldView-2 imagery, Maxar, copyright 2024, U.S. Government Plus license. TerraSAR-X copyright 2019 German Aerospace Center (DLR); courtesy of Simon Plank, DLR.

Using airborne and satellite remote sensing to rapidly construct DEMs is critical for measuring topographic changes caused by deposition of newly erupted material at an erupting volcano. Satellite stereophotogrammetry (WorldView) and SAR (TanDEM-X) can capture high-resolution (<5 m) terrain data, which have been used for such applications as tracking island growth during the 2016–17 eruption of Bogoslof Island (Coombs and others, 2019) and measuring lava effusion rates at Kīlauea (Poland, 2014). SfM software permits in-house construction of submeter resolution DEMs from aerial images captured during routine overflights, including FLIR imagery where volcanic plumes can obscure visible imagery or where eruptive activity is difficult to discern optically (for example, Patrick and others, 2017).

Aerial surveys using visible and infrared photogrammetry and single-pass InSAR (NASA’s GLISTIN-A) offer higher resolution (<5 m) and on-demand DEM generation capabilities. Additionally, overflights with crewed aircraft and UAS provide airborne video and imagery surveys that can be integrated with image processing techniques for quantifying eruption dynamics and constructing orthorectified visible and thermal maps at submeter resolution (Patrick and others, 2017; Neal and others, 2019; Dietterich and others, 2021). The USGS Hawaiian Volcano Observatory has routinely acquired repeat thermal imagery from aircraft overflights to provide situational awareness to the National Park Service, county, State, and Federal emergency response managers, and the public. This technique was critical to mapping lava flow development and associated hazards during Kīlauea’s 2018 lower East Rift Zone eruption (fig. G3).

Eruption rates can be determined by comparison of topography over time, which can be derived from both optical



Base image from Worldview 2, May 23, 2018
Maxar, copyright 2024, U.S. Government Plus license

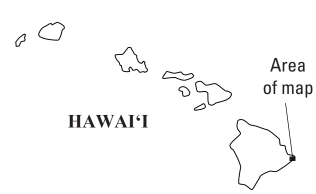


Figure G3. Flow field thermal imagery from June 27, 2018, overlain on a preeruptive image of the 2018 lower East Rift Zone eruption of Kīlauea, Hawaiʻi, acquired on May 23, 2018. Photograph shows a U.S. Geological Survey Hawaiian Volcano Observatory geologist using a forward looking infrared camera from a helicopter to thermally image a lava flow at Kīlauea, Hawaiʻi. Photograph by T. Orr, U.S. Geological Survey, March 4, 2016.

(for example, Bagnardi and others, 2016) and SAR (for example, Poland, 2014) satellite data, but even simple morphological changes detected qualitatively in remote sensing imagery can provide important hazards information. The successful forecast of hazardous eruptive activity at Merapi, Indonesia, in 2010, for instance, was based on SAR amplitude imagery that was used to characterize lava dome growth patterns over time (Pallister and others, 2013). Effusion rates calculated using airborne thermal-imaging cameras can be more accurate than those determined from satellite because of the higher dynamic range and spatial resolution of the aerial observations (Patrick and others, 2017). SAR and photogrammetric techniques both produce digital surface models that may not represent the bare-earth topography in vegetated areas; therefore, airborne lidar data are essential for capturing topographic change with high accuracy and resolution within vegetated areas. Ground-based multiview stereo imagery, InSAR, and lidar can similarly track local topographic change, such as from lava flow emplacement, mass movements, and crater floor and lava lake level changes.

Recommendations

Baseline observational monitoring of levels 1 and 2 volcanoes is met by collecting satellite imagery (see “Collect Baseline Imagery and Topography” section). For baseline observational monitoring of level 3 volcanoes, we suggest deployment of at least one telemetered optical camera. For level 4 volcanoes, we suggest at least two telemetered cameras with different views of the volcano. At level 4 and frequently restless volcanoes, we also suggest regular (about monthly) acquisition of high-spatial-resolution, georeferenced and orthorectified visible, short-wave infrared (SWIR), and radar satellite images or aerial imagery.

During periods of unrest or eruption, imagery and DEMs would be acquired more frequently from satellite or aerial platforms, satellite or airborne SAR, or lidar surveys. Visual and (or) thermal orthomosaics could be constructed from airborne imagery surveys using SfM techniques. Additional telemetered and (or) time-lapse or video and video-lapse camera capabilities could be deployed if practical to document lava extrusion, pyroclastic flows, and other eruptive phenomena. Acquisition intervals of ground-based cameras could be adjusted to capture rapidly changing activity.

Detect and Characterize Thermal Emissions Through Time

Prior to eruption, as magma intrudes a volcano, surface heat flow may increase, resulting in the development of, or changes to existing, fumarolic activity, fractures, hot springs, crater lakes, and snow and ice fields. Detection and tracking of these changes relative to the volcano’s background state can play an important role in understanding the processes involved in volcanic unrest. Regular monitoring of thermal emissions can therefore offer a valuable tool to detect unrest at an early stage, and likely well before an eruption occurs. For example, long-term temperature changes have been tracked prior to eruptions using ground-based radiometers (Harris and others, 2005b), satellite imagery (Reath and others, 2019; Girona and others, 2021), and airborne thermal data (Schneider and others, 2008; Wessels and others, 2013).

During eruptions, characterization of volcanic radiative power is an important parameter that can be used to evaluate the magnitude and evolution of volcanic activity. Tracking total volcanic radiative power through time can reveal precursory thermal signals and how eruptive activity is evolving through time (for example, Wright and others, 2002; Coppola and others, 2016). Radiative power can be estimated from low-spatial-resolution data but requires knowledge of the subpixel fraction of the hot feature—information obtainable from higher resolution satellite data or thermal-imaging radiometric data. The volcanic radiative power can be further refined into estimates of effusion rate (Harris and others, 1997) and total eruption volume, which are important for understanding magma supply and producing lava flow hazard assessments. Although effusion rate can be calculated by using low-spatial-resolution satellite data (Harris and others, 1997; Patrick and others, 2016), more accurate estimates are obtainable by using data from thermal-imaging cameras, which have a much higher dynamic range and spatial resolution (Harris and others, 2005a; Patrick and others, 2017; fig. G3). These higher resolution data collected from the ground or from aircraft can be combined with the synoptic, repetitive view that satellite data provide to produce a time series of lava effusion over the course of an eruption.

Instrumentation

Infrared data track changes in the locations, sizes, and strengths of thermal signals through time and can be used to detect unrest and measure eruption rates. Imagery can be acquired at various spatial and temporal resolutions—meteorological satellites capture low spatial but high temporal resolution with Advanced Very High Resolution Radiometer (AVHRR), GOES, Moderate Resolution Imaging Spectroradiometer (MODIS), and VIIRS for hot objects (Harris and others, 1997; Dehn and others, 2002; Wright and others, 2002, 2004); less frequent medium-resolution imagery from the Advanced Spaceborne Thermal Emission and Reflection Radiometer (ASTER) and Landsat (Flynn and others, 2001; Ramsey and Dehn, 2004); and very high spatial and temporal resolution from infrared cameras in aircraft- and ground-based applications (Harris and others, 2005b; Vaughan and others, 2005; Patrick and others, 2014, 2017). These sensors also vary in spectral bands and resolution. Hot surfaces can also be detected and characterized in the near-infrared and SWIR wavelengths by more sensors and at higher spatial resolution than thermal infrared at all scales (for example, Landsat 8 and 9 thermal infrared resolution is 100 m, but its SWIR resolution is 30 m).

Automated hot spot detection and processing of thermal emissions from frequent low- and moderate-resolution satellite data, as well as infrared web cameras, can provide critical early detection and characterization of volcanic unrest and eruption. Algorithms for automated thermal time series processing of satellite imagery detect thermal signals above background at volcanoes and may be used to generate alerts of new activity and quantify thermal output through time during an eruption at the spatial and temporal resolutions of the satellite data (for example, MODVOLC or MIROVA; Wright and others, 2004; Coppola and others, 2016). Quantitative thermal flux (volcanic radiative power; fig. G4) can importantly be used to estimate lava effusion rates through time (for example, Harris and others, 2005a; Loewen and others, 2021). Similarly, automated processing of thermal web camera imagery allows detection of sudden increases in temperature indicative of eruption onset, as well as tracking and characterization of thermal features (for example, Patrick and others, 2014). These and other algorithms may also detect weaker thermal emissions prior to eruption (Dehn and others, 2000; Girona and others, 2021).

Detection, precise mapping, and analysis of subtle changes in thermal features must utilize higher resolution airborne and satellite imagery. Signs of unrest may be expressed in moderate- to high-resolution thermal imagery, such as changes in geothermal areas (for example, Vaughan and others, 2020). During eruption, high-resolution thermal imagery can track activity in detail, such as lava lake level or lava flow or dome extents, lava transport systems such as channels and tubes, surface temperatures, and flow dynamics (for example, Wessels and others 2013; Patrick and others, 2014, 2017; figs. G2 and G3). Telemetered thermal infrared camera data can provide the most frequent views, whereas high-resolution airborne and satellite imagery have lower temporal resolution.

Recommendations

Any available low-spatial-resolution satellite data may be used to establish baseline thermal features and for automated hot spot detection and alerting for anomalous significant thermal activity for levels 1 and 2 volcanoes (for example, Coppola and others, 2016).

For levels 3 and 4 volcanoes, it is important to frequently characterize volcanic thermal flux (for example, radiative power) to detect eruption precursors and to track changes in thermal features over time. We suggest acquiring baseline knowledge of the distribution and temperature of thermal features. Mapping and temporal comparisons can be made at moderate to high resolution with ASTER and Landsat 8 or 9, tasking WorldView-3 SWIR and (or) nighttime Landsat 8 or 9, and thermal imagery captured during repeat airborne (both crewed and unoccupied) and ground-based missions (for example, Vaughan and others, 2012). We also suggest using high temporal, low spatial resolution meteorological satellite data for automated hot spot detection, tracking, and alerting for anomalous significant thermal activity (for example, Coppola and others, 2016). During eruptive activity, it is important to distinguish lava effusion from other thermal signals, such as

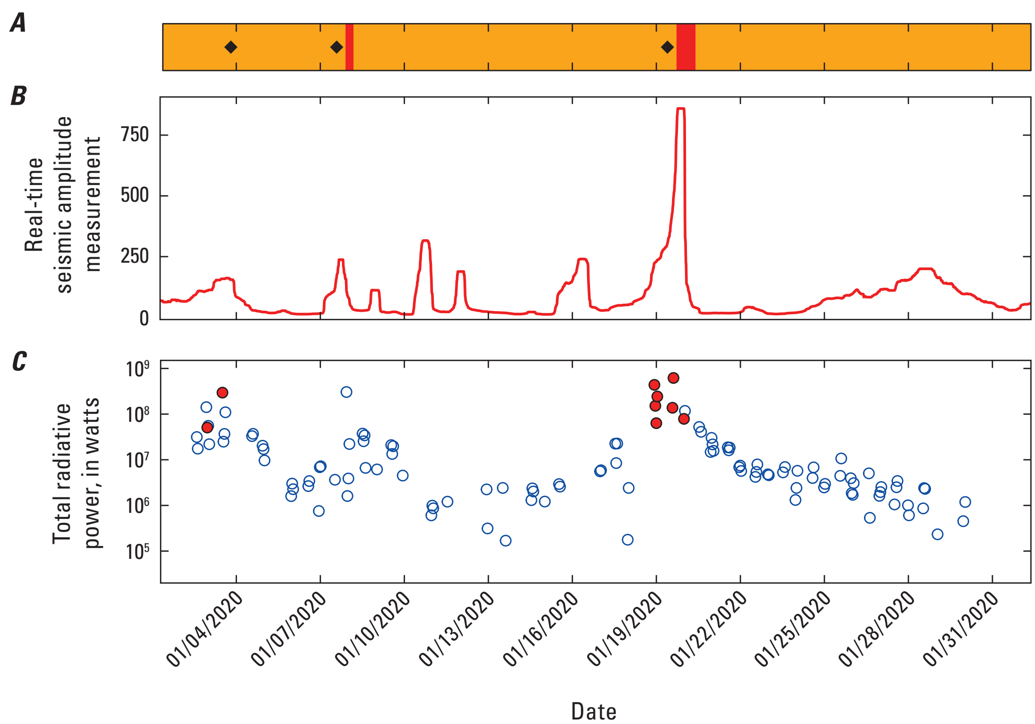


Figure G4. Plots showing the timeline of eruptive activity at Shishaldin Volcano, Alaska, in January 2020. *A*, Changes in aviation color code. Black diamonds indicate when ash-producing events occurred. *B*, Seismic tremor, shown as real-time seismic amplitude measurements (Endo and Murray, 1991) at station SSBA, plotted as a 12-hour average. *C*, Quantitative radiative power from the Visible Infrared Imaging Radiometer Suite (VIIRS) (following method described by Loewen and others, 2021). Open circles show when elevated surface temperatures were detected; filled circles show when those elevated temperatures exceeded the VIIRS sensor range and the pixels were saturated. The offset in ash events from color code changes is because of retrospective timing of the event starting date compared to when formal notifications or color code changes were issued. Dates shown as month/day/year.

pyroclastic flows and Strombolian activity, and to characterize a time series of thermal flux (and estimated eruption rate) from lava domes, flows, and other features. High-spatial-resolution georeferenced mapping of lava dome and flow extents, lava channel and lava tube systems, and thermal signatures that capture the style and level of activity through time can be done using commercial satellite SWIR, and (or) airborne or ground-based thermal cameras (for example, fig. G2).

At level 4 volcanoes, we additionally suggest using higher resolution airborne and ground-based thermal imagery for on-demand and high-temporal-resolution observations. We also suggest deployment of at least one telemetered thermal infrared camera, and (or) low-light camera, ideally co-located with an optical camera, where allowed by environmental conditions and viewing geometry, and additionally warranted by activity level (for example, areas of frequent eruptive activity, or where there is a history of thermal precursors prior to eruptions).

During periods of unrest and eruption, additional cameras (telemetered and [or] time-lapse or video or video-lapse) with low-light or thermal-infrared capabilities could be deployed to further document lava extrusion, fountaining and Strombolian activity, pyroclastic flows, and other phenomena that have thermal signatures. Throughout an eruption, it is important to distinguish between these phenomena and to characterize a time series of thermal flux (and estimated eruption rate) from lava domes, flows, and other features. High-spatial-resolution georeferenced mapping of lava dome and flow extents, channel and tube systems, and thermal signatures that capture the style and level of activity through time could be done using commercial satellite SWIR and ground-based and (or) airborne thermal cameras (for example, Patrick and others, 2017). A very high-temporal-resolution time series of volcanic radiative power can be based on GOES imagery acquired every 1 to 10 minutes to capture rapid changes in eruption dynamics.

Summary—Recommendations for Levels 1–4 Networks

Level 1.—Collect moderate-resolution baseline imagery and at least 10-m-resolution DEMs to provide a synoptic overview of volcanic landforms, deposits, and thermal features for later comparison. Implement automated hot spot detection and alerting.

Level 2.—Same as for level 1.

Level 3.—Collect and regularly update high-resolution baseline optical and thermal imagery and at least 5-m-resolution DEMs to provide a synoptic overview of volcanic landforms, deposits, and thermal features for later comparison and for detailed geologic and hazard mapping; deploy one telemetered ground-based optical camera. During periods of unrest or eruption, construct high-resolution visual and (or) thermal orthomosaics and generate new DEMs as necessary, perform automated time-series processing and mapping of thermal features, and implement automated hot spot detection to generate alerts of anomalous thermal activity.

Level 4.—Same as level 3, except baseline DEMs would ideally be at least 1-m resolution to improve hazards modeling; deploy two telemetered ground-based cameras and a thermal camera where appropriate. Regularly acquire high-resolution stereopairs and orthorectified optical and SWIR satellite and (or) aerial imagery. During periods of unrest or eruption, frequently characterize thermal flux with satellite, airborne, and ground-based thermal imagery to track changes and estimate eruption rates, increase acquisition of imagery and DEMs, deploy additional telemetered and (or) time-lapse cameras with video, low-light, or thermal-infrared capabilities.

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